

UNIVERSITÀ
DEGLI STUDI
DELL'AQUILA

Lesson 7: Trees and Ensembles

Technologies for Artificial Intelligence

Fabio Antonini, Università degli Studi dell'Aquila

20 November 2026

Agenda

- A fourth family, built on a threshold
- Decision trees, and why one alone is mediocre
- Depth as the bias-variance dial, again
- Bagging and random forests
- Feature importance, and its limits
- Gradient boosting
- Choosing among trees, forests and boosting

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Exercise 6, before we start

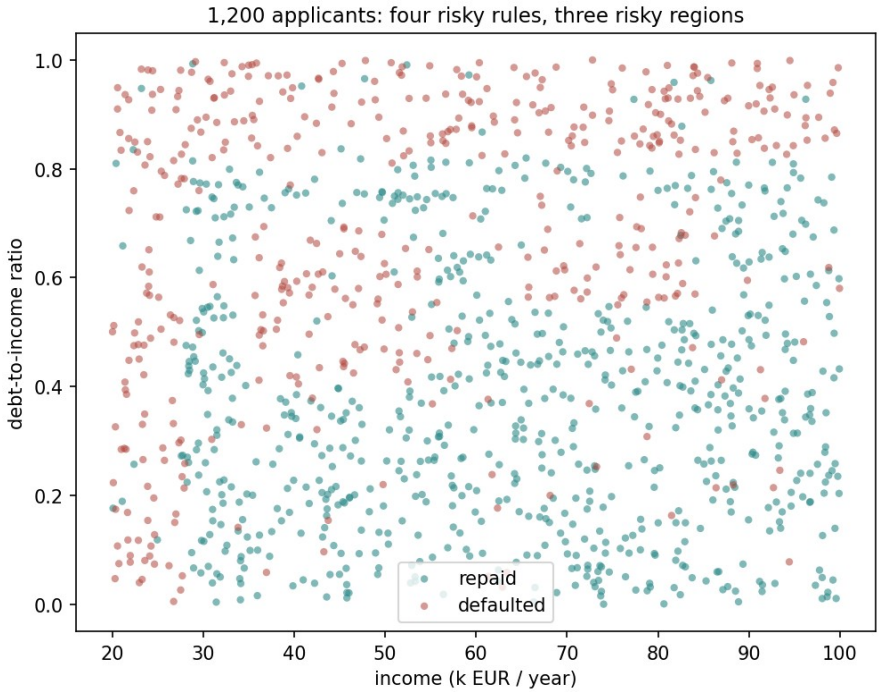
- What counts is the **defence of which family**, not the winning score
- The gap to watch for: a strong accuracy number with no check of *why* the assumption behind it held
- Today’s models come with the same obligation

1,200 loan applicants, two figures each

- Income in thousands of euros a year, and debt-to-income ratio
- Underwriting is a **checklist**: an income floor, a debt ceiling, and two “stressed” combinations, each figure ordinary alone but risky together
- **7%** of labels flipped: no honest model beats about **0.93**, the **noise ceiling**
- **38.7%** defaulted, so the majority baseline is **0.613**

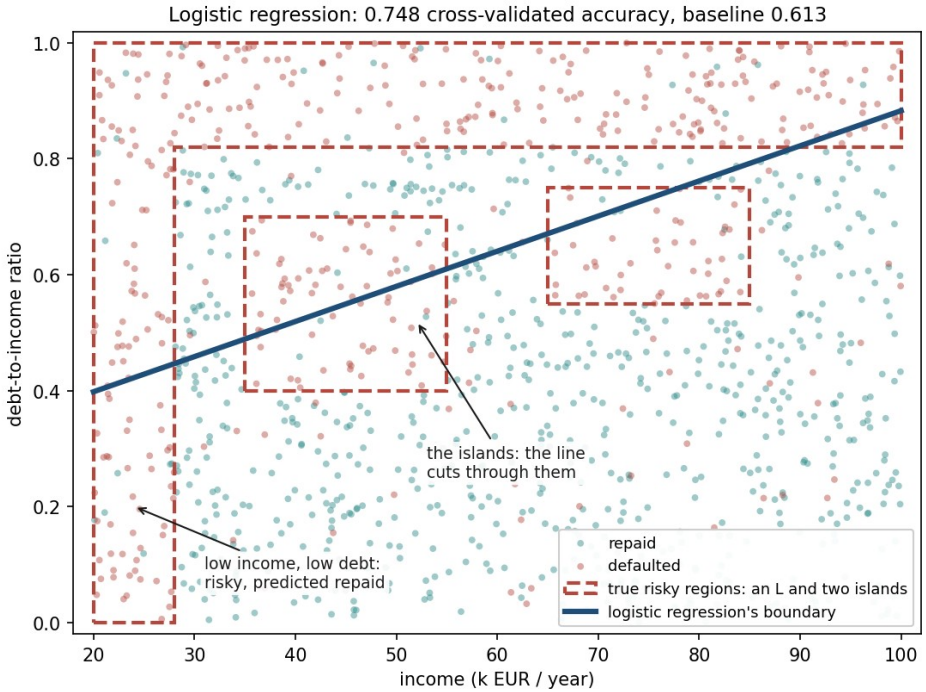
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Four risky rules, three risky regions



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

No straight line separates them



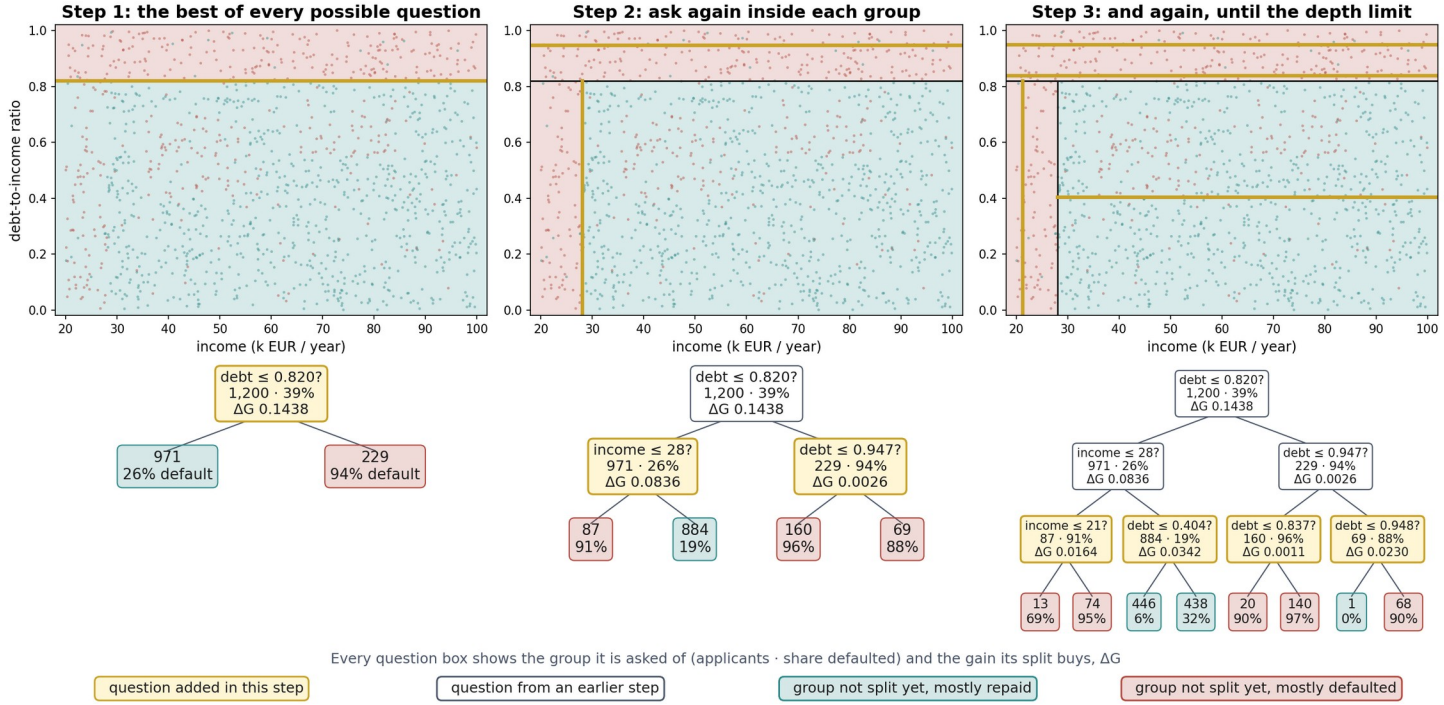
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The whole algorithm, in one sentence

- Try **every feature and every threshold**; keep the split leaving the **purest** two groups
- Apply it; repeat on each group, independently
- Stop at a depth limit, a pure group, or too few examples to split
- No coefficients, no gradient: thresholds are found by **searching**
- **Greedy**: whichever split helps most *right now*, with no lookahead

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The algorithm, drawn



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

How pure is a group of examples?

- A leaf that is all one class needs no more questions
- A group half one class, half the other, is the worst case: a coin flip
- **Impurity** is a number that captures this: 0 for a pure group, highest for an even mix. “As pure as possible” then means “with the lowest impurity”

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Gini impurity

$$G = 1 - \sum_{c=1}^{C} p_c^2$$

p_c : share of class c (defaulted, repaid)

slide 8's 229 : $1 - 0.939^2 - 0.061^2 = 0.115$

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

For two classes, one number

- G = 2p(1-p), where p is the share defaulted and 1-p the share repaid
- **0** when the group is pure (p = 0 or p = 1)
- **0.5**, the maximum, at p = 0.5
- On the full 1,200 applicants, p = 0.387 gives G = 0.474

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The gain a split buys

- A split divides a parent P into a left child L and a right child R; *m* counts each one's examples: at the root, 1,200 in P = 971 in L + 229 in R
- Worth taking if the children's impurities, **averaged**, are lower than the parent's: at the root 0.474 against 0.330
- The average is **weighted by size**: one example cut off alone would halve a plain average, and barely moves a weighted one

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The reduction in impurity

$$\Delta G = G_P - \left(\frac{m_L}{m_P} G_L + \frac{m_R}{m_P} G_R \right)$$

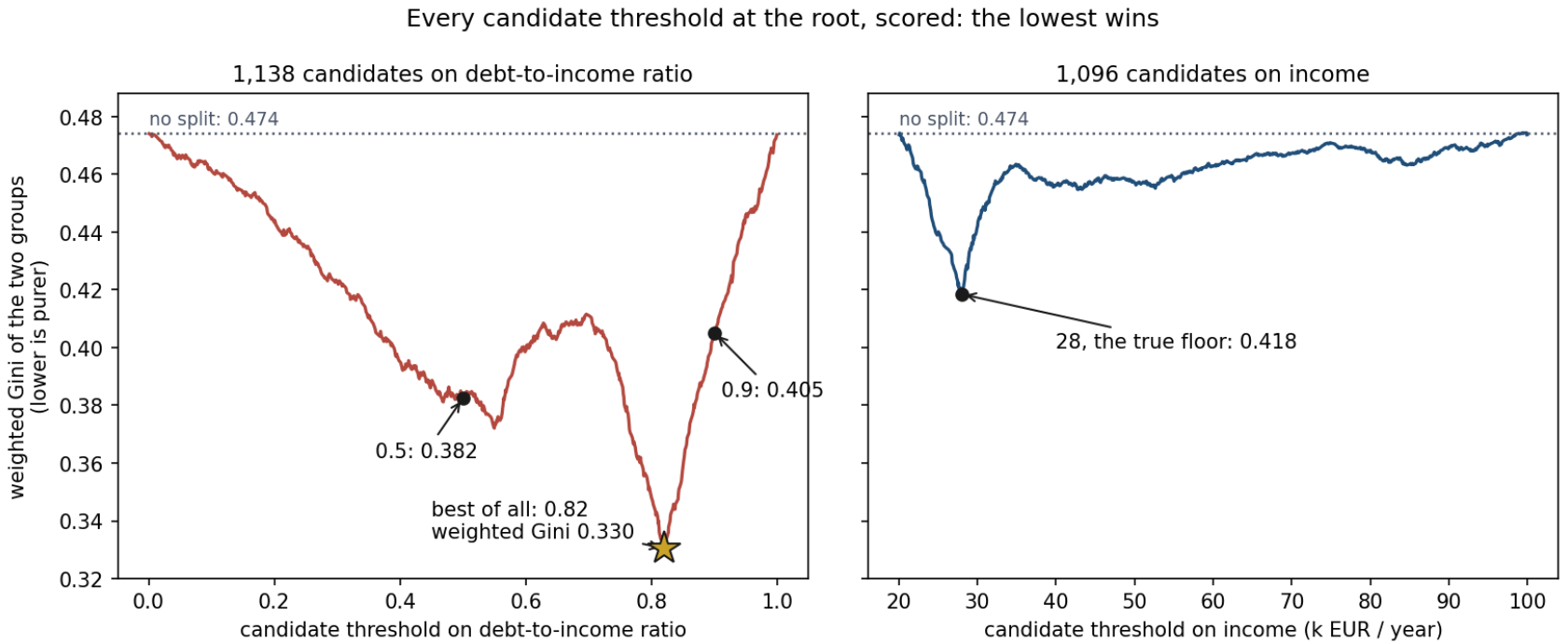
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Scanning for the best split

- Sort a feature’s values: every **midpoint** between neighbours is a candidate threshold
- Infinitely many thresholds, but **finitely many splits**: any threshold between the same two neighbours makes the same two groups
- Score every candidate on **every feature**; keep the best pair, then repeat inside each child
- **Greedy**: ΔG is judged one split at a time, never jointly

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Every threshold, scored



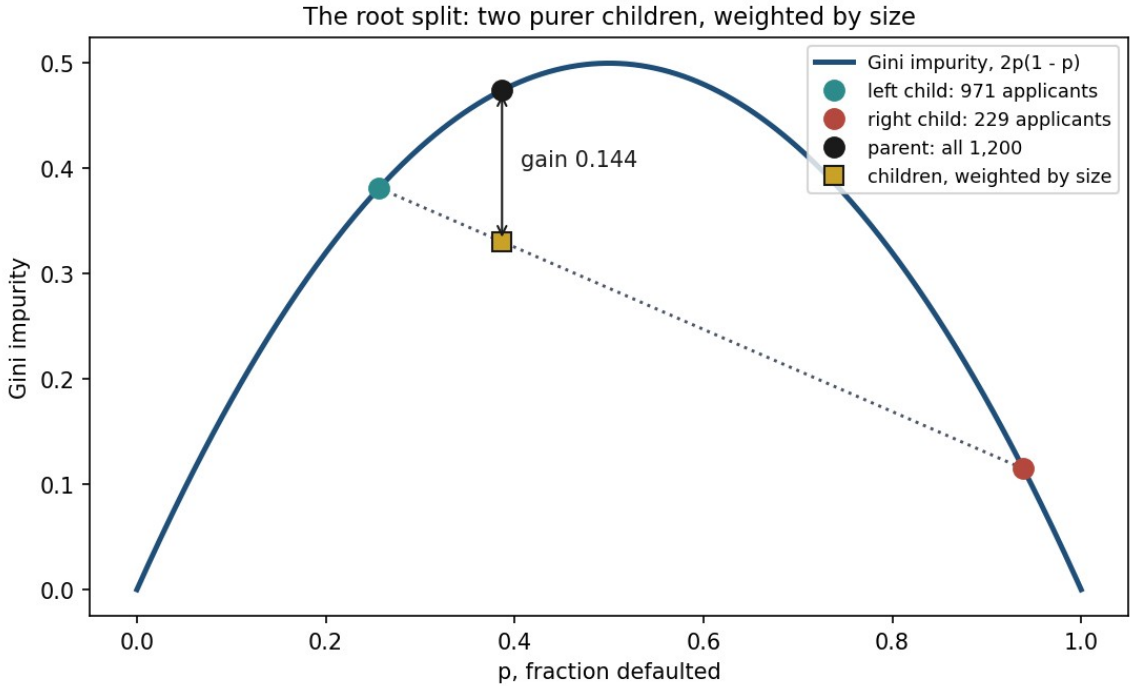
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Worked example: the root split

- Parent: 464 of 1,200 defaulted, G = **0.474**; the search finds `debt_ratio <= 0.82`, the **debt ceiling**
- Left: 249 of 971 defaulted, p = 0.256, G = 2 × 0.256 × 0.744 = **0.381**
- Right: 215 of 229, p = 0.939, G = 2 × 0.939 × 0.061 = **0.115**
- Weighted: 0.809 × 0.381 + 0.191 × 0.115 = 0.330; gain 0.474 - 0.330 = **0.144**

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The root split, on the Gini curve



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The loop, in seven lines

```
def grow(g):
  if stop(g): return leaf(g)
  gain = {c: ΔG(g, c) for c in splits(g)}
  best = max(gain, key=gain.get)
  if gain[best] <= 0: return leaf(g)
  L, R = split(g, best)
  return node(best, grow(L), grow(R))
```


UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Checked against scikit-learn

- The from-scratch implementation and scikit-learn's `DecisionTreeClassifier` make **the same prediction for all 1,200 applicants**, at every depth tested
- Not a coincidence: both search the same space by the same greedy rule
- From here on, the notebooks use scikit-learn's implementation: the same algorithm, compiled, not a different one

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

What greedy misses

- The search never asks “which pair of splits helps most together”: only “which single split helps most right now”
- A feature that only pays off **in combination** with a later split can be invisible to a shallow tree
- Exactly what happens to this dataset’s two interior “stressed” islands, covered next

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Depth is the bias-variance dial

- `max_depth`: the most questions **one applicant** answers
- **Shallow tree**: few questions, only the coarsest structure. High bias, low variance
- **Deep tree**: enough questions to isolate single points. Training accuracy **exactly 1.000**, even on data with no structure, unless two examples share every feature value but not the label

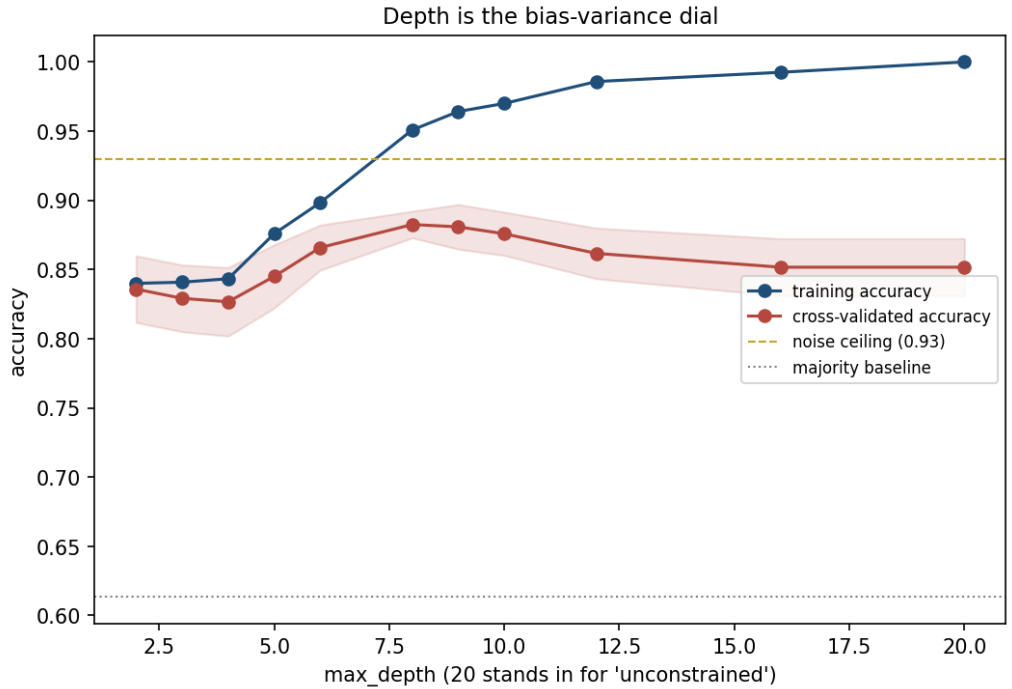
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Measured, at seven depths

max_depth	Training accuracy	Cross-validated, 5 folds (mean ± standard deviation)
2	0.840	0.836 ± 0.024
4	0.843	0.827 ± 0.025
6	0.898	0.866 ± 0.016
8	0.951	**0.882 ± 0.010**
12	0.986	0.862 ± 0.018
16	0.993	0.852 ± 0.021
unconstrained	**1.000**	0.852 ± 0.021

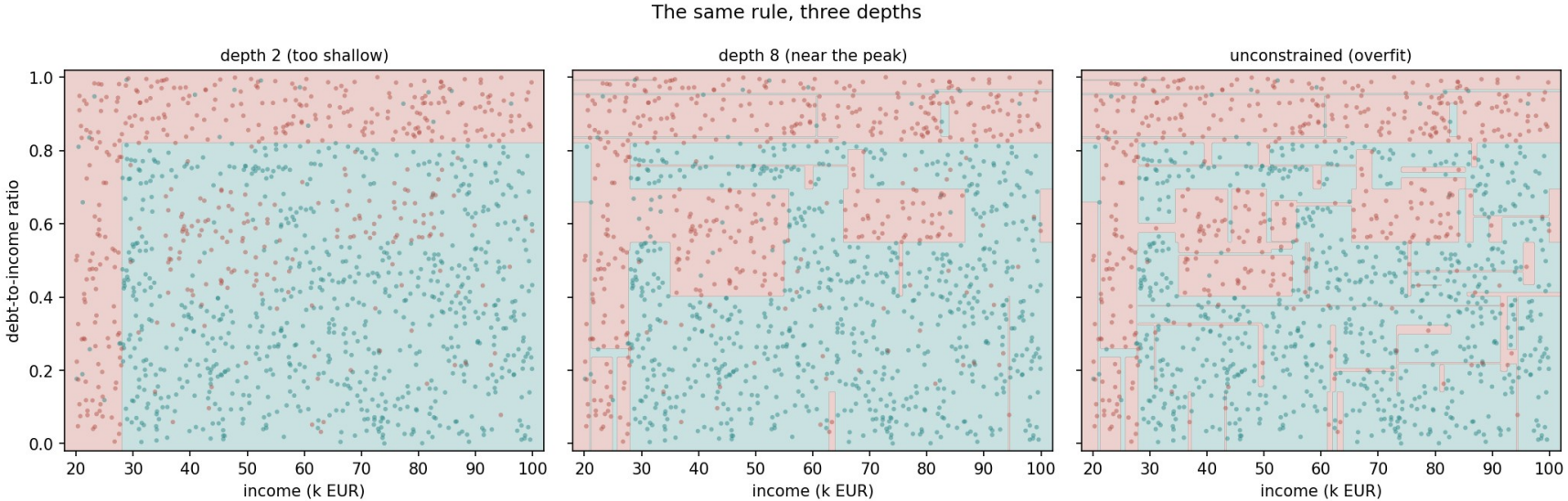
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The curve bends downward



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

The same rule, three depths



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Reading a tree, and the knob that keeps it small

- `max_depth` **caps every path**: each node above the cap is split if it can be, worth it or not
- `max_leaf_nodes` is a **budget of leaves**, spent on the largest ΔG first
- `max_leaf_nodes = 9`: **0.908** training, **0.901** cross-validated, above depth 8's 0.882; thresholds 28 and 0.82
- In code: `export_text(tree)` prints the tree as if/else rules

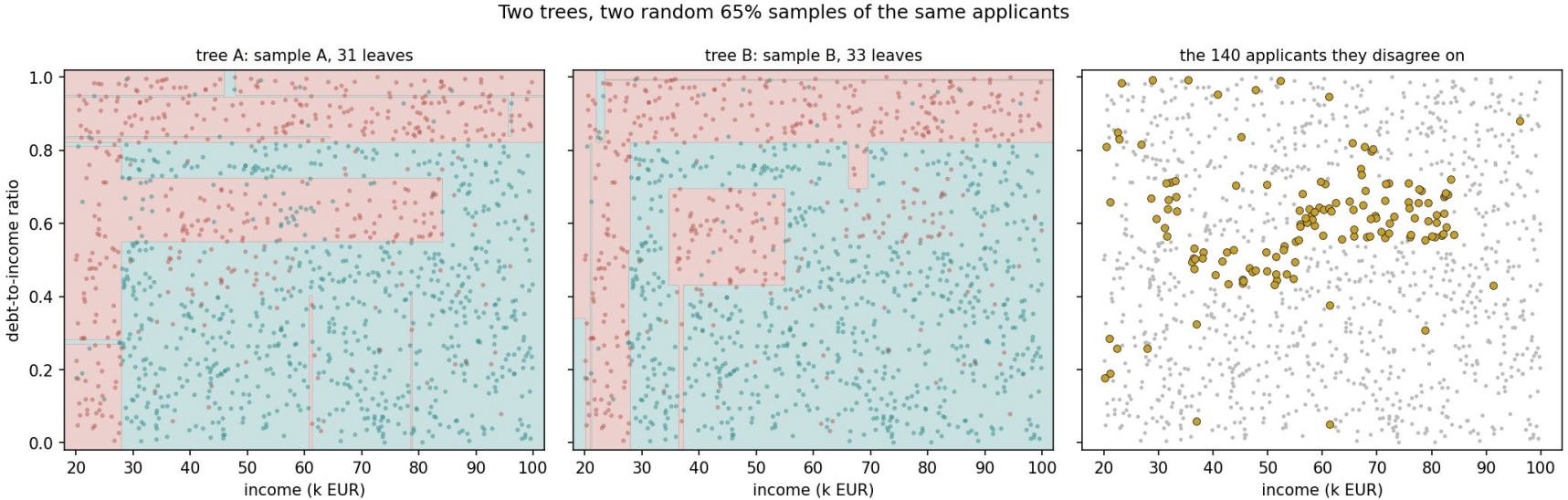
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

A tree that doesn’t stay still

- Two trees at `max_depth = 6`, each on 780 of the 1,200 applicants, drawn **without replacement** (no one twice)
- The root, decided on all 780, agrees to within **0.001**: the debt ceiling
- The 29 or so questions below it, each on fewer applicants, do not: **31 and 33 leaves**, **opposite verdicts for 140** of the 1,200 (11.7%)

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Two trees, 140 different verdicts



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Notebook 1, live

section, and its slides	you call (→ the number to come back with)
1. look at it first, 4-6	`make_pipeline(StandardScaler(), LogisticRegression())` in `cross_val_score(..., cv=StratifiedKFold(5))` → 0.613 against 0.748
2. a tree, split by split, 7-20	your own `gini` and `best_split`, then `DecisionTreeClassifier(max_depth=d)` → 100.0% identical predictions; gain 0.1438
3. the depth dial, 21-24	`DecisionTreeClassifier(max_depth=d)` in `cross_val_score` → peak 0.882 at depth 8
4. what a tree costs, 25-27	`DecisionTreeClassifier(max_leaf_nodes=9)`, `export_text` → 0.908 training, 0.901 cross-validated; two trees, 140 opposite verdicts

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Break

- Twelve minutes

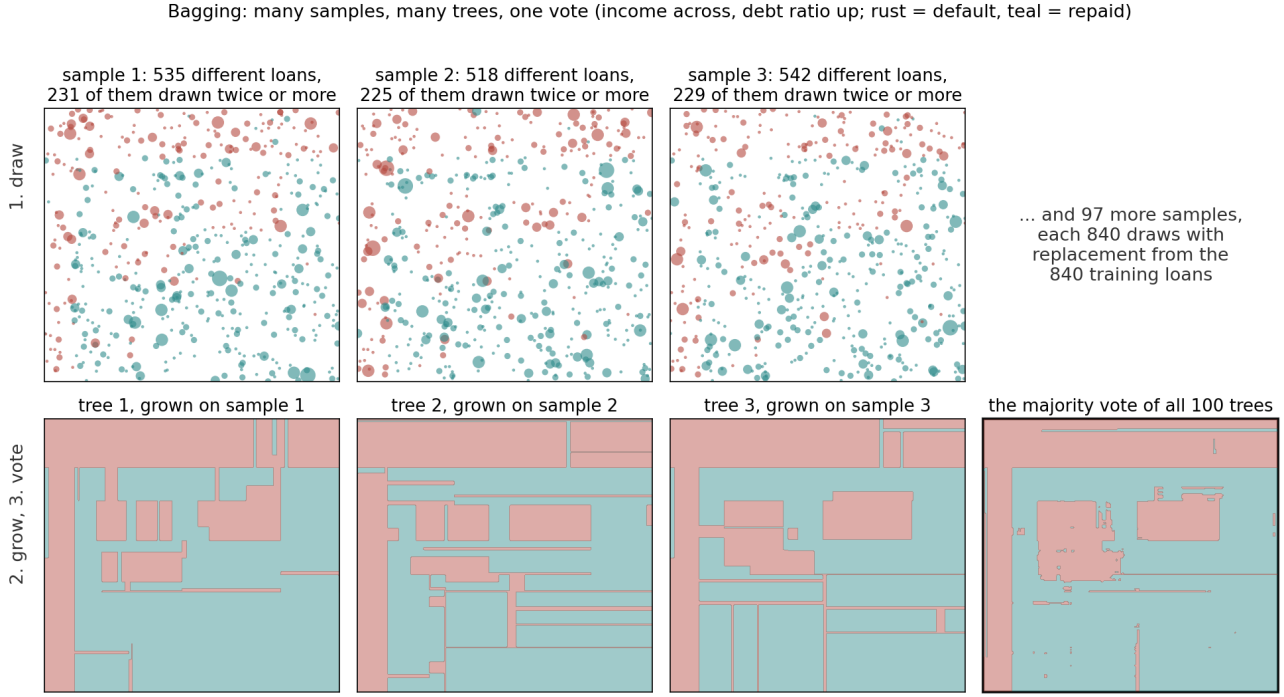
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Bagging: averaging away the instability

- Slide 27: two trees disagree on 140 of 1,200, so **many trees, combined by a vote**, should cancel what **differs from tree to tree**
- **Bootstrap sample**: m rows drawn **with replacement** from the m **training** rows: some twice, some never
- One **fully grown** tree per sample, `n_estimators` times, then a **majority vote**: **bagging** (bootstrap aggregating)

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Bagging, drawn



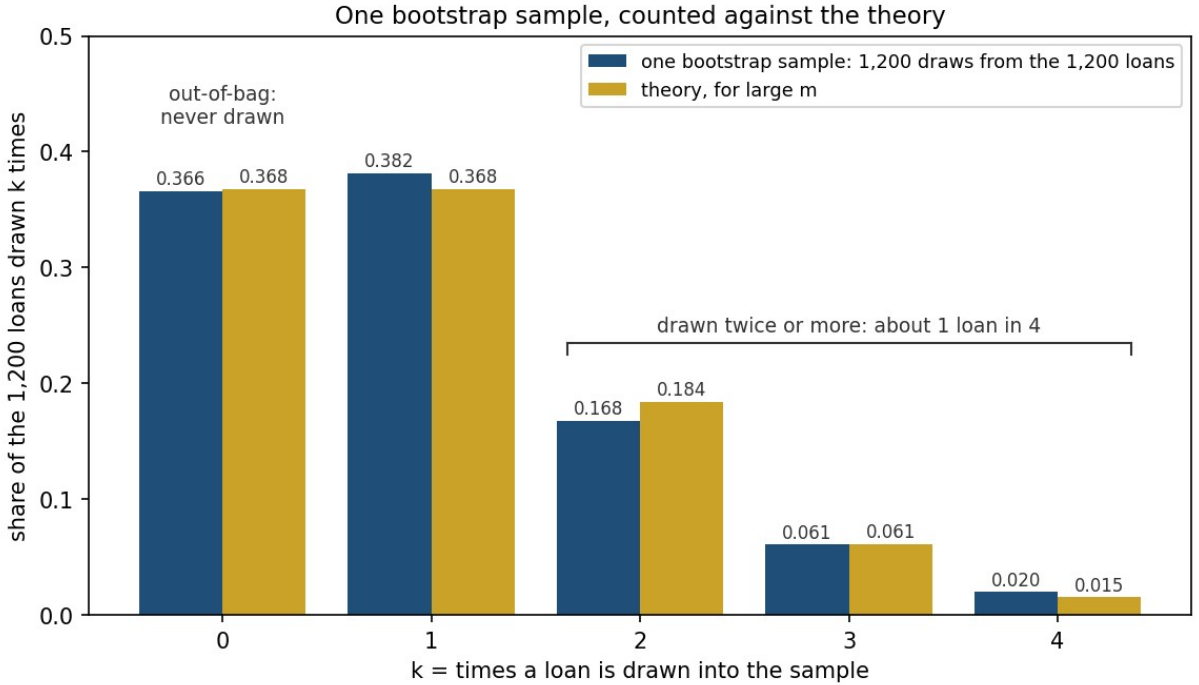
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

How much data one bootstrap sample sees

- One draw misses a row with probability 1 - 1/m; m independent draws all miss it with $(1 - 1/m)^m$, which tends to $e^{-1} \approx 0.368$
- So about **63.2%** of the distinct rows are drawn; the other **36.8%** are **out-of-bag (OOB)** for that tree
- Still m draws: at m = 1,200, on average 1,200 × (1 - 0.3677) = 759 distinct rows, 441 repeats

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

One bootstrap sample, counted against the theory



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Out-of-bag: each loan judged by the trees that never saw it

Out-of-bag: each loan is judged only by the trees that never saw it

	tree 1	tree 2	tree 3	tree 4	tree 5	tree 6	tree 7	tree 8	tree 9	tree 10	out-of-bag verdict	true label	
loan 0	seen ×1	seen ×2	seen ×4	seen ×1	seen ×1	seen ×1	seen ×1	seen ×1	**D**	seen ×1	**D**	**D**	**right**
loan 3	seen ×1	seen ×4	**R**	seen ×2	**R**	**R**	**R**	seen ×2	seen ×2	**R**	**R**	**D**	**wrong**
loan 4	seen ×1	seen ×1	seen ×1	seen ×2	seen ×3	**D**	seen ×3	seen ×1	seen ×1	seen ×2	**D**	**D**	**right**
loan 5	seen ×2	**R**	seen ×1	seen ×2	seen ×2	seen ×1	**R**	**R**	seen ×2	seen ×2	**R**	**R**	**right**
loan 6	seen ×4	seen ×2	seen ×1	seen ×4	seen ×1	**D**	seen ×1	**D**	**D**	seen ×3	**D**	**D**	**right**
loan 8	**R**	seen ×1	**R**	seen ×2	seen ×1	**R**	**R**	**D**	seen ×2	seen ×2	**R**	**R**	**right**
loan 10	**R**	**R**	**R**	seen ×2	**R**	**R**	seen ×2	seen ×1	seen ×2	seen ×4	**R**	**R**	**right**
loan 11	**D**	**D**	seen ×1	**D**	seen ×1	seen ×2	**D**	**D**	**D**	**D**	**D**	**D**	**right**

grey: the loan is in that tree's sample (×2 = twice), so that tree does not vote on it; gold: the tree never saw it and votes - D defaulted, R repaid

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Out-of-bag against cross-validation

- All 1,200 loans, **no split**; 300 bagged trees, `oob_score=True`
- **Out-of-bag**: each loan judged by the trees that never saw it (about 110): **1,090 right, 0.9083**
- **5-fold cross-validation**: five ensembles on 960, each scored on the other 240: **1,083 right, 0.9025** (± 0.021)
- Same question; out-of-bag needs **no split, no second fit**

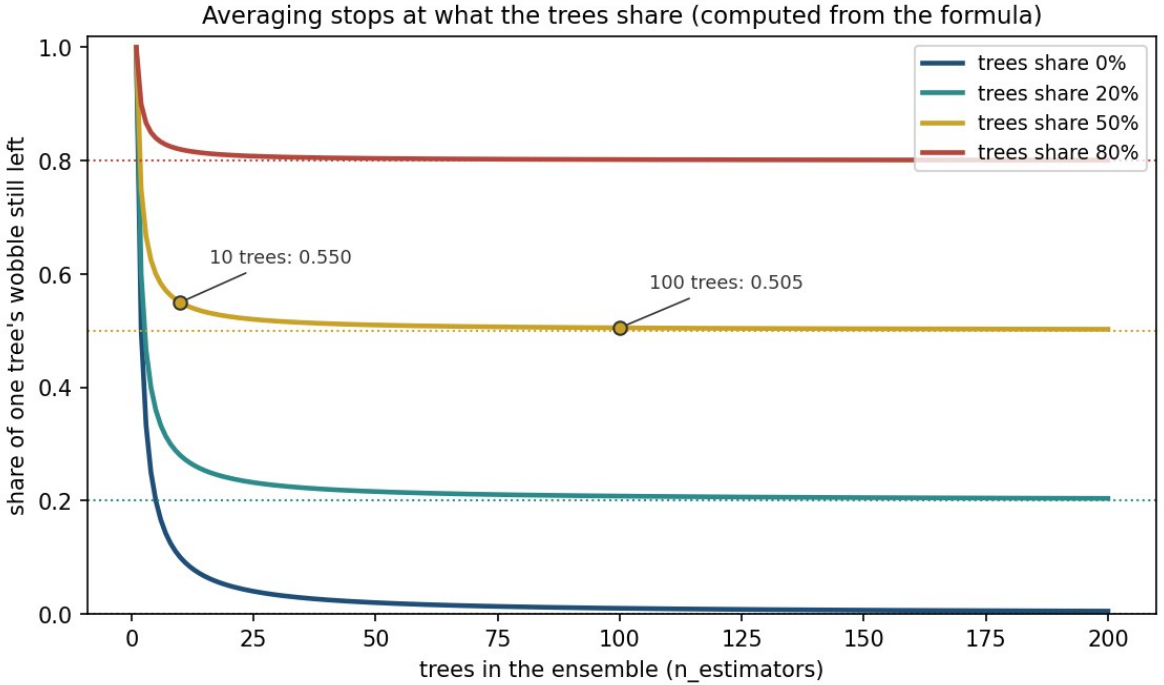
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Why averaging helps, and where it stops

- Each tree’s verdict depends on its **bootstrap luck**: averaging cancels it
- All trees grow from the **same 1,200 loans**: what they learn from them is **shared**, and does not cancel
- Ten experts reading one newspaper: their quirks cancel, its mistakes do not
- So averaging stops at a **floor**; lower needs trees **less alike**

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Averaging stops at what the trees share

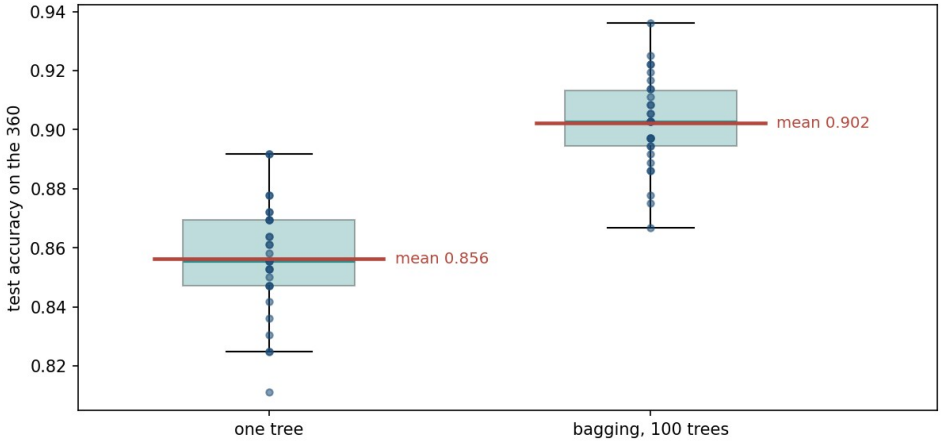


UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Bagging against one tree, on 30 splits



repeated with 30 different random splits: 30 test accuracies per model



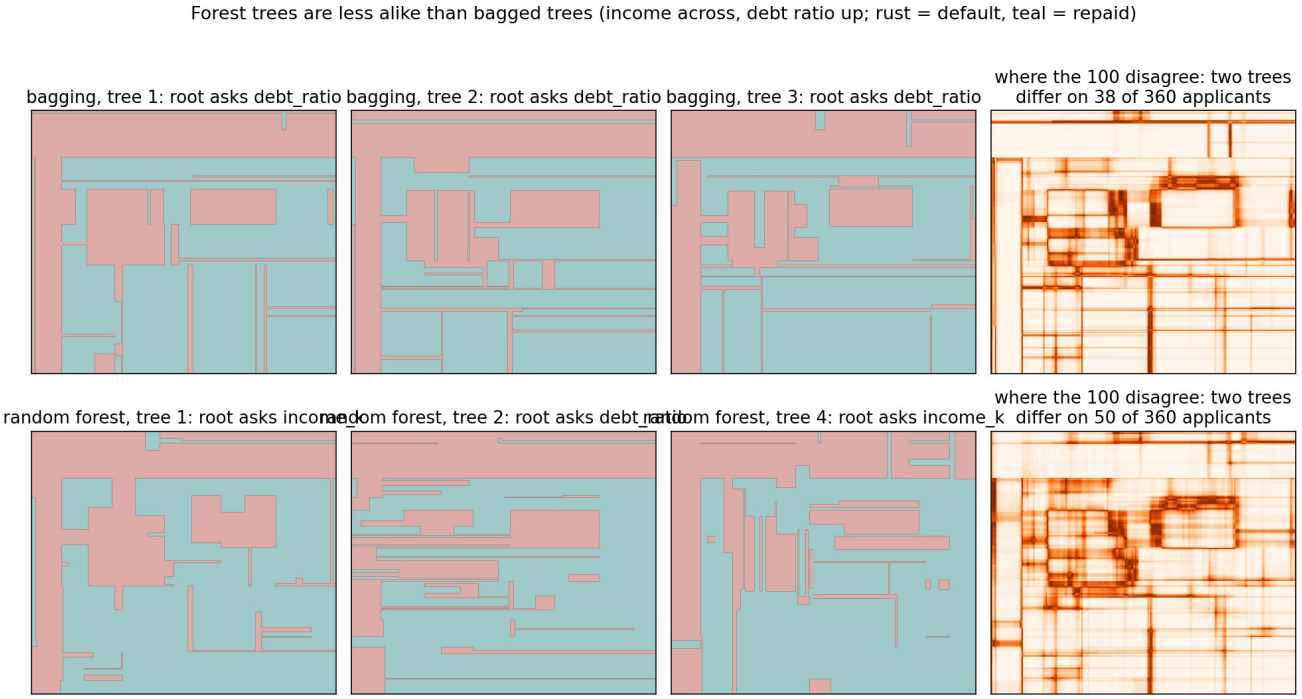
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Random forests: making the trees less alike

- Bagging: every split sees **every feature**; all 100 trees open on `debt_ratio`
- **Random forest**, `RandomForestClassifier()`: each split sees only `max_features` random columns (√n, rounded down), redrawn at every split; here n = 2, so **one**
- Roots: 51 on debt, 49 on income; a tree **worse** alone (0.828 against 0.837), the forest **better**

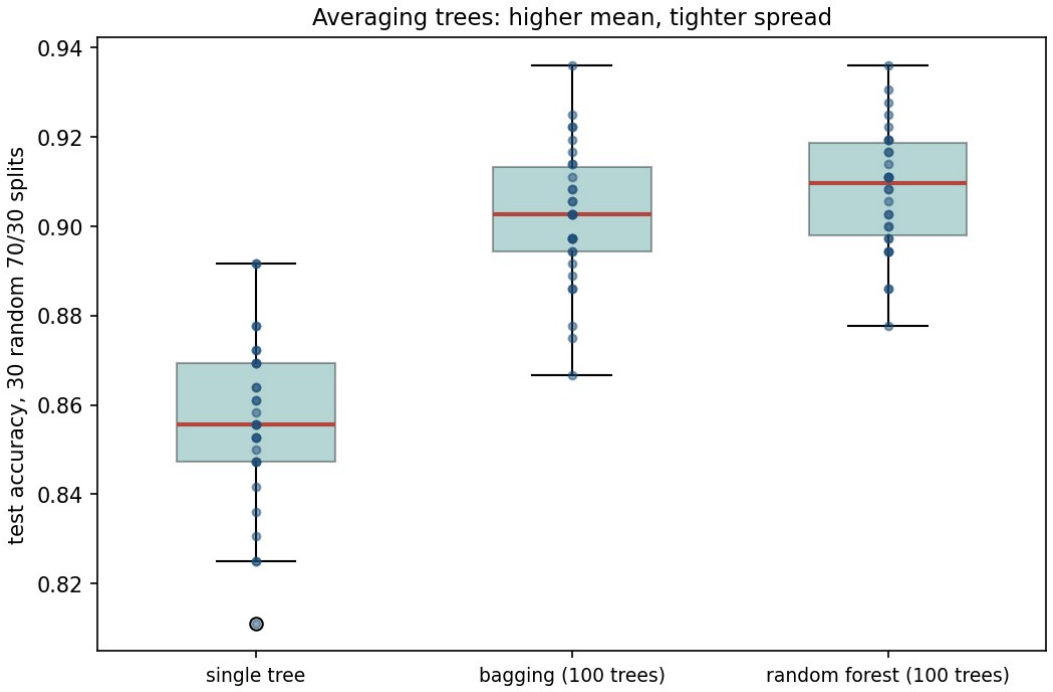
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Bagged trees against forest trees



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Random forests, measured



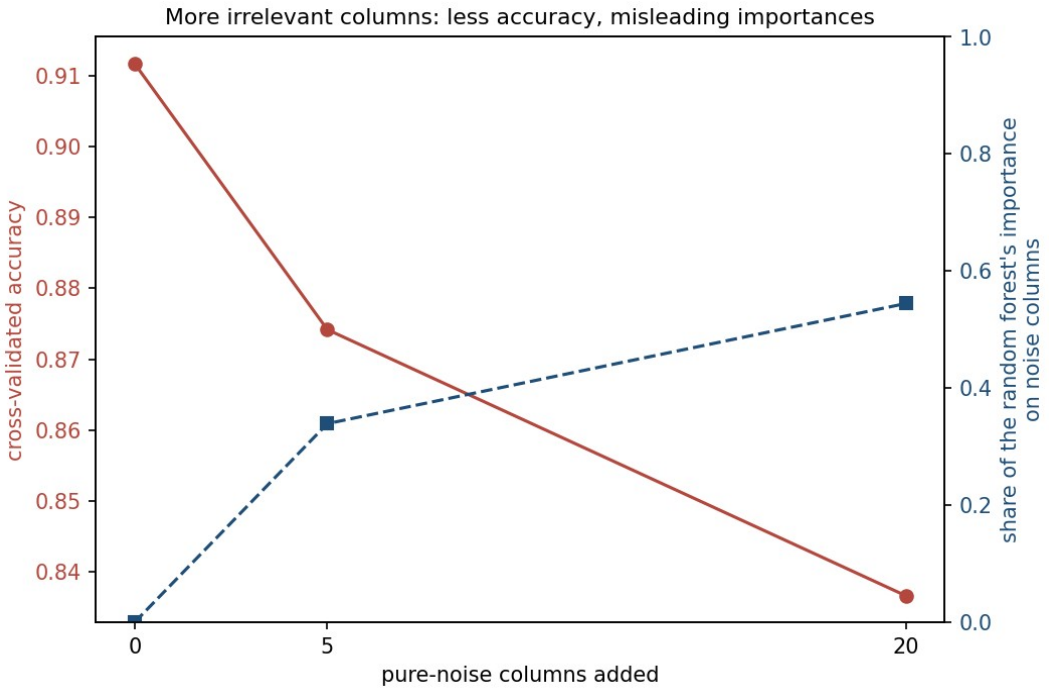
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Feature importance: what the forest claims to tell you

- **Feature importance**: the impurity reduction credited to each feature over every tree and split, scaled to add up to 1
- With only the two real features here, it names exactly those two
- What happens once we add columns of random numbers, unrelated to default?
- In code: `forest.feature_importances_`

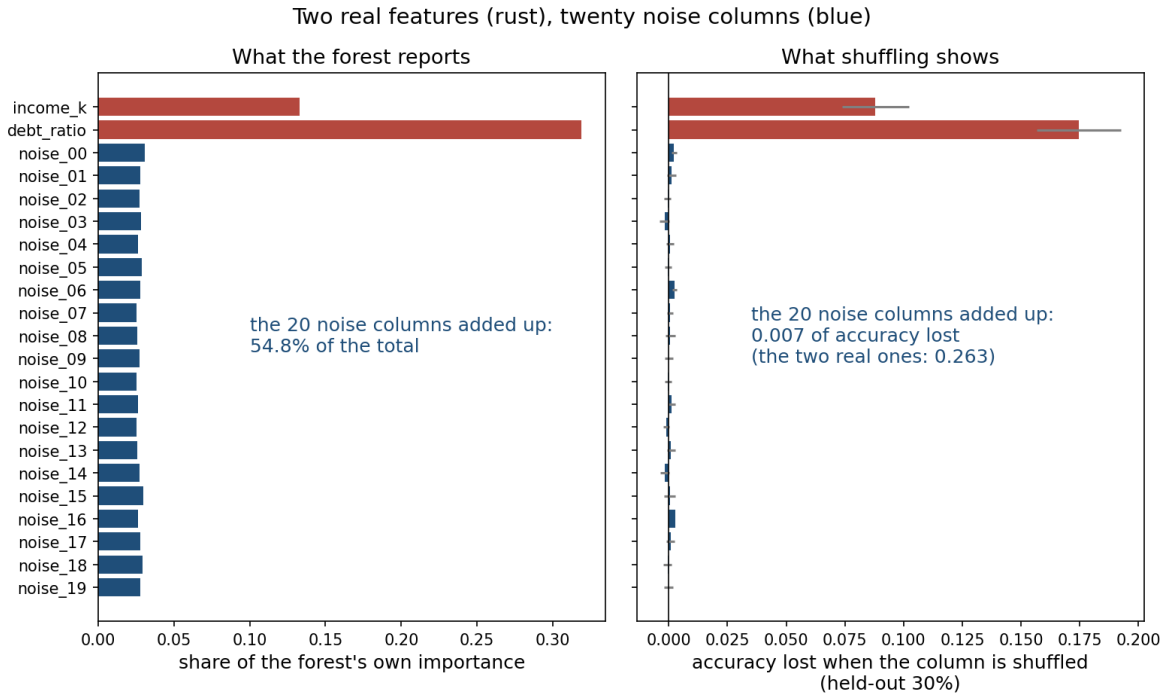
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

More columns, more noise



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Permutation importance



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Notebook 2, live

section, and its slides	you call (→ the number to come back with)
1. bagging from scratch, 30-31	`train_test_split`, your bagging against `BaggingClassifier`, `accuracy_score` → 0.872 both; 1 of 29 flipped right
2. why averaging works, 32-37	`oob_score=True`: `.oob_score_` against `cross_val_score` → 0.9083, 0.9025
3. variance, measured, 38, 41	tree, bagging, `RandomForestClassifier`, 30 splits → 0.856, 0.902, 0.908
4. what a forest adds, 39-40	`.estimators_`, `tree_.feature[0]` → roots 100, 51 and 49
5. feature importance, 42-43	`.feature_importances_`, 0 to 20 noise columns → 54.5% on noise
6. checking the importances, 44	`permutation_importance`, test split → real 0.09 and 0.18, noise ≈ 0

Gradient boosting: correcting mistakes in sequence

- Bagging and random forests build trees **independently**, then average
- **Boosting** builds them **in sequence**: each new tree corrects what the trees so far got wrong
- Each is a shallow **regression tree**, a **weak learner**: it predicts a number, a correction, not a class
- The learning rate α sets **how much** of each correction is applied

Building the ensemble, one tree at a time, for squared error

$$F_0(x) = \bar{y}, \qquad F_t(x) = F_{t-1}(x) + \alpha \cdot h_t(x)$$

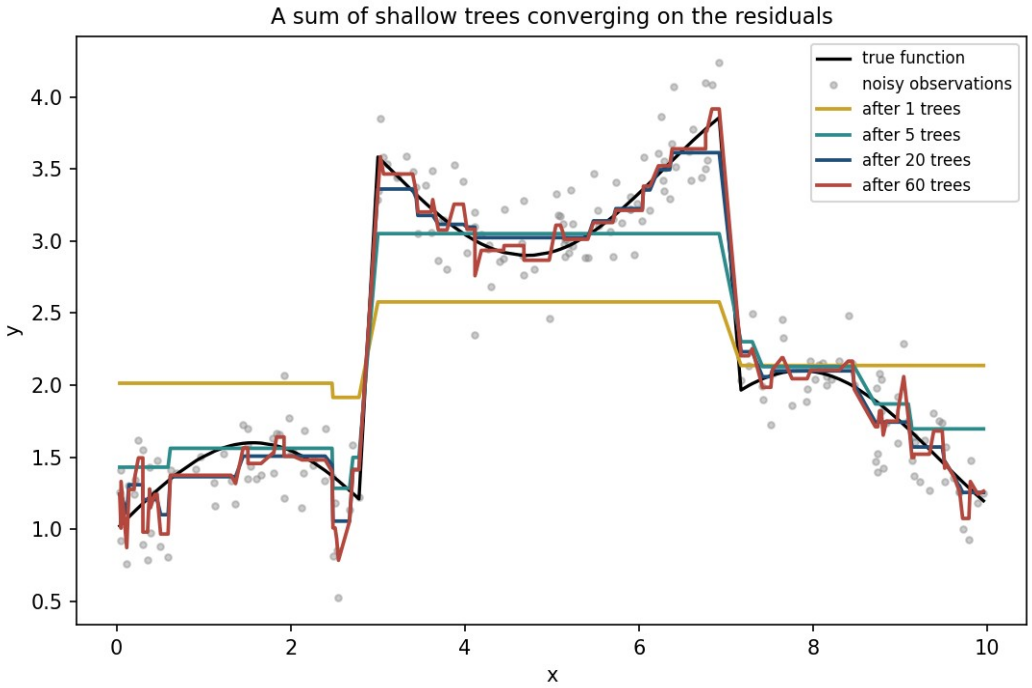
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

What each tree is fit to

- For squared error, what remains unexplained is the **residual**, $y - F_{t-1}(x)$
- In general, **functional gradient descent**: h_t approximates the negative gradient of the loss with respect to F_{t-1}
- With $L = \frac{1}{2}(y - F)^2$, $-\partial L/\partial F = y - F$: the two coincide
- In code: `DecisionTreeRegressor(max_depth=2)`, fit to `y - prediction`

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

A step plus a sine wave, fit by 60 shallow trees



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Reading the picture

- **1 tree**: mean squared error (MSE) against the true function 0.395, barely moves the flat guess
- **5 trees**: MSE 0.068, the coarse shape appears
- **20 trees**: MSE **0.012**, the ensemble’s best point
- **60 trees**: MSE rises back to 0.018, tracking noise, not signal

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Classification: the same idea, a different gradient

- y is 0 or 1; the score F becomes a probability through lesson 4’s logistic function, $p = \sigma(F)$ (this σ is not the σ^2 of the variance slides)
- The loss is lesson 4’s **log-loss**
- Each tree fits the **pseudo-residual** y - p, the negative gradient: lesson 4’s error term
- Start: p = 0.387 for all, so the first tree fits +0.613 to defaulters and −0.387 to the rest

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

On the loan data

- `GradientBoostingClassifier`, defaults: **100 trees**, learning rate $\alpha = 0.1$
- Reaches **0.902 ± 0.020** cross-validated accuracy
- Already ahead of any single tree from earlier in the lesson

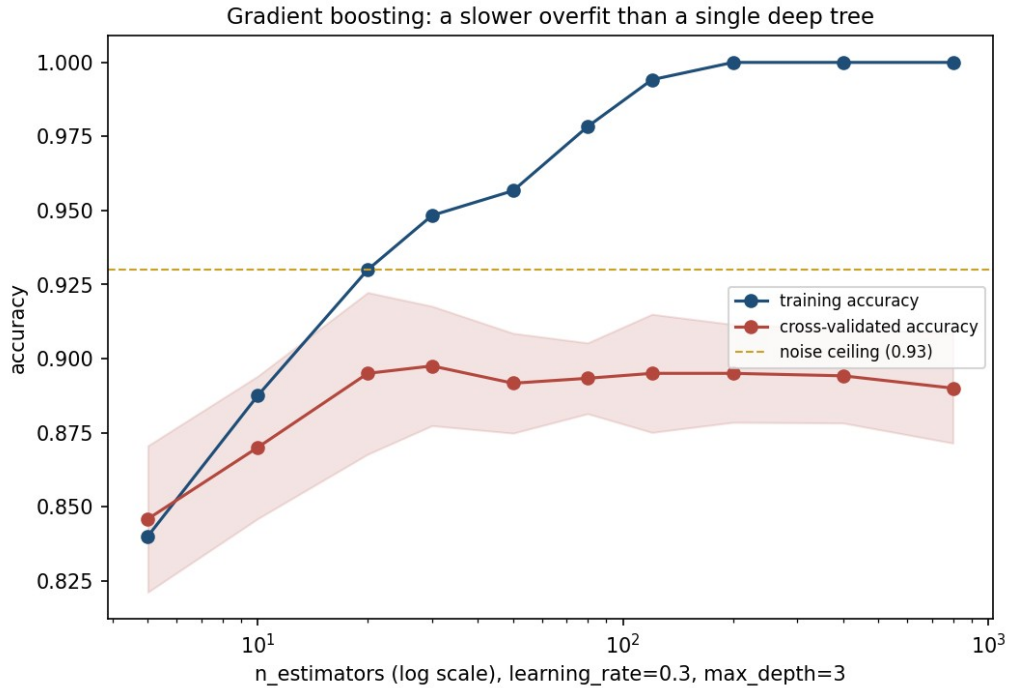
UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Boosting never stops fitting on its own

`n_estimators` (α = 0.3, depth 3)	Training accuracy	Cross-validated accuracy
5	0.840	0.846 ± 0.025
10	0.888	0.870 ± 0.024
20	0.930	0.895 ± 0.027
30	0.948	**0.897 ± 0.020**
80	0.978	0.893 ± 0.012
200	1.000	0.895 ± 0.017
800	1.000	0.890 ± 0.019

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Training climbs to 1.000 and stays; the honest curve turns over first



UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Fixing 120 trees, varying only the learning rate

Learning rate α	Training accuracy	Cross-validated accuracy
0.02	0.888	0.872 ± 0.025
0.05	0.914	0.897 ± 0.022
0.10	0.939	**0.902 ± 0.022**
0.30	0.994	0.895 ± 0.020
1.00	1.000	0.878 ± 0.020

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Notebook 3, live

section, and its slides	you call (→ the number to come back with)
1. boosting from scratch, 46-50	`DecisionTreeRegressor(max_depth=2)` on the residuals, 60 times → MSE 0.0119 at 20 trees
2. classification, 51-52	`GradientBoostingClassifier()` in `cross_val_score` → F0 = −0.461; 0.902
3. more trees, 53-54	the same, `learning_rate=0.3`, `n_estimators` from 5 to 800 → peak 0.897 at 30
4. learning rate, 55	the same with `n_estimators=120` and `learning_rate` from 0.02 to 1.0 → best 0.902 at 0.1
5. leaderboard, 57	every model in one `cross_val_score` loop → random forest 0.911

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Every method, on the same five folds

Model	Cross-validated accuracy
Majority baseline	0.613
Logistic regression	0.748
Single tree, unconstrained	0.852
Single tree, `max_depth = 8`	0.882
Gradient boosting, 30 trees	0.897
Single tree, `max_leaf_nodes = 9`	0.901
Bagging, 100 trees	0.904
Random forest, 100 trees	**0.911**
noise ceiling	*≈ 0.93*

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Three findings

- **On this dataset**, the ensembles land within 1.5 points of each other: **whether you ensemble** matters more than **which**
- A **nine-leaf tree** reaches 0.901, a point from bagging: on rectangles a small tree is enough, and a person can read it
- The **unconstrained tree is the worst**, though the most flexible

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

How to choose, in practice

Situation	Reach for
The decision must be explained to the person it affects	A single tree, small enough to read; if even that is too opaque, none of these models
Tabular data, accuracy matters most, no explanation required	A random forest: the best score here, at its default settings
Accuracy matters, and there is time to tune the number of trees and the learning rate together	Gradient boosting, tuned on validation data
Many columns, few of them informative	Any tree method, but check its importances by permutation
Too few rows to set a validation set aside	A random forest, judged by its out-of-bag score

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

54% of the importance landed on noise

- 20 noise columns, 2 real ones: not small, not a bug
- **The menu**: 4 of 22 columns per split; in two splits of three (0.662), no real feature
- **Small groups**: with all 22 on offer, noise still takes **28%**, from deep splits
- The check: **permutation importance**, slide 44

What to take away

- A tree splits greedily on Gini impurity, matching scikit-learn
- Depth is the bias-variance dial: accuracy peaked at **depth 8**
- Bootstrap resampling leaves **36.8%** of rows out-of-bag: free validation
- **54% of a random forest’s importance landed on 20 noise columns**, today’s number

UNIVERSITÀ DEGLI STUDI DELL’AQUILA

Homework

- **Exercise 7**, discussed at the start of **Friday 27 November**
- `Exercises/07_trees_and_ensembles.md`

UNIVERSITÀ DEGLI STUDI DELL'AQUILA